

Municipal Spending in Spain: Spatial Approach

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Abstract: This paper provides, for the first time, an insight into the influence of urban, political, demographic, and economic factors on several categories of municipal spending, after controlling for the spatial effects. The models of spatial two-step least squares (S2SLS), spatial lag, and spatial error are applied to a sample of 3,204 Spanish municipalities for the year 2005. The results show that the spending of a municipality is influenced by its neighboring municipalities' spending. Concerning geographical factors, population density has a negative impact on municipal expenditures per capita. The negative impact of population density on per capita total expenditures has policy relevance because it indicates that planning decisions aiming at increase population density will reduce per capita government expenditures, both in small and larger cities. Political ideology does not influence spending. Nevertheless, strong political governments do spend more. Therefore, it is not that the incumbents belong to left or right, but their strong majority which allows them to carry out their policies by spending more. The policy issue at stake here is whether municipal political majorities should be tempered, so that opposition surveillance keeps spending under control. DOI: 10.1061/(ASCE)UP.1943-5444.0000138. © 2013 American Society of Civil Engineers.

CE Database subject headings: Financial factors; Local governments; Political factors; Urban development; Spain.

Author keywords: Financial factors; Local governments; Political factors; Urban development.

Introduction

As Alt and Lowry (1994) suggest, fiscal outcomes result from the interaction of economic and political circumstances at the subnational level: it is clear that economic factors are not enough to explain public sector financial performance, but rather social and political frameworks play a role (Borge 1995; Seitz 2000). There is a rich body of literature evaluating the impact of these political, social, geographical and economic factors on municipal spending. In the last years, a methodology (spatial econometrics), which controls for the impact of geography on municipal behavior, is being applied. This methodology assumes that a municipality is influenced by its neighboring municipalities in several economic, demographic and social aspects.

To evaluate these relationships, we have obtained data on 3,204 Spanish local governments (LG) for the year 2005. This sample enables us to assess which factors explain LG spending, from a spatial econometric approach.

We think the interest of this paper is twofold. On the one hand, the empirical evidence of the influence of geographical, political, demographic and economic factors on LG spending is ambiguous.

Therefore, this paper provides further insight into the LG spending after controlling for the spatial effects. On the other hand, the Spanish transition to democracy in the 1970's led to the decentralization of the Central Government in favor of the 17 Spanish Regional Governments. However, many political experts claim that a "second decentralization" is needed, where the municipalities receive enough financial resources to fund all the duties that, de facto and de jure, they carry out. Consequently, it is important to have an in-depth knowledge about the spending behavior of LG in Spain.

The paper is organized as follows. We first briefly describe local government in Spain. Then we review the literature on municipal spending. We continue with a brief outline of Spanish LG. After describing the methodology, regressions results are shown. Finally, empirical results are discussed and our conclusions are outlined.

Local Government in Spain

The Spanish public sector is divided into three levels: the State, 17 Regional Governments (which are called Autonomous Communities in Spain) and LG (50 provinces and 8,112 municipalities). 84% of municipalities have less than 5,000 inhabitants, with only 13% of the total Spanish population living in them. 60% of municipalities have less than 1,000 inhabitants. There are 15 big cities in Spain (over 250,000 inhabitants).

Each municipality has a mayor, a cabinet, and a professional administration. The mayor is the head of the executive, and is elected indirectly by the citizens. The electoral system is based on proportional representation. Municipal elections of over 66,000 aldermen are held every four years.

The political parties at the Spanish national level also operate in LG. Thus, there are two main national parties at the local level: the Socialist Party (progressive) and the Popular Party (conservative). These two parties account for 66.36% of the total number of aldermen in our sample, becoming the most important parties at the municipal level.

The Spanish constitution lays down two basic principles with regard to LG: financial autonomy and financial sufficiency. The

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Note. This manuscript was submitted on June 6, 2011; approved on August 30, 2012; published online on September 3, 2012. Discussion period open until November 1, 2013; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Urban Planning and Development*, Vol. 139, No. 2, June 1, 2013. © ASCE, ISSN 0733-9488/2013/2-79-93/\$25.00.

first principle means that municipalities, provinces and islands can make their own decisions on resources and spending. The principle of financial sufficiency is aimed at guaranteeing that all municipalities have the necessary resources to exercise their duties. In order to be financially sufficient, LG finances rely on the following resources: income from self-owned assets, local taxes, surcharges on the Regional Governments, shares in State and Regional Governments' taxes, subsidies, public fares, credit transactions, fines, and other sanctions. Municipalities are allowed to both raise local taxes and charge tariffs for the services they provide.

The municipal expenditures are classified according to three criteria: organizational, economic and functional. The organizational classification discloses who spends. The economic classification reports the economic nature of the spending. Finally, the functional classification details the purpose of the expenditure.

This paper bases its analysis on the functional classification of expenditures, taking into account the highest level of disaggregation available. This classification is very interesting from a dual perspective. For the government, it indicates the LG investment priorities at every moment, and from the perspective of citizens it allows considering whether these priorities really reflect society's needs.

Within the scope of this work, we use seven dependent variables: total expenditure and six categories of expenditures (solid waste, water, housing, security, education and culture).

Solid waste (*solidwasteh*) and water (*waterh*) include water sanitation and distribution and waste disposal, respectively. Housing (*housingh*) includes services related to housing and urban development, as well as the complementary of these (construction, improvement and maintenance of housing and shelters, land acquisition, planning and regime of the urban soil, buildings, lighting and parks and gardens). Security (*securityh*) means services related to safety and mobility of citizens and those the LG must provide, in cooperation with the State or the Region to support services for protecting and defending civil actions in case of nature calamities, extinguishing fires, and actions designed to protect the LG assets. Finally, education (*educationh*) and culture (*cultureh*) include the creation, maintenance and operation of schools and educational institutions or libraries, museums and archives, respectively.

Municipalities should provide a series of basic services depending on their population:

- All municipalities: public lighting, cemeteries, waste collection, public cleaning, drinking water supply, sewer system, access to urban areas, road surfacing, and food and drink control.
- Municipalities with more than 5,000 inhabitants: public parks, public libraries, market, and waste management.
- Municipalities with more than 20,000 inhabitants: civil defence, social work, fire safety, and sports facilities.
- Municipalities with more than 50,000 inhabitants: public transport and environment protection.

Literature on Municipal Spending

Urbanization/Geography

The theoretical and empirical literature has investigated the impact of population density and population growth on public expenditures.

Since the early work of Fabricant (1952), urbanization (proportion of citizens living in urban areas) and population density became classical variables in explaining LG fiscal behavior. Furthermore, classic theory of public finance shows that urbanization

determines government growth in the states, and even more, in the local authorities. The explanation of this long standing theory, known as Wagner's Law, is the pressure for social progress and the resulting changes in the relative proportions of private and public economy (Wagner 1958).

From a theoretical point of view, the population density is expected to impact local spending. Bails and Tieslau (2000) suggest two ways in which population density can be related to the level of state and local government spending. On the one hand, a dense population may increase the marginal benefits of spending if it creates unique public goods problems: a positive relationship between density and per capita expenditure is assumed. On the other hand, population density may lead to economies of scale in the provision of LG services. In this latter case, we expect a negative relationship between these two variables.

The empirical evaluation of this theoretical assessment shows mixed results. Some authors find that public expenditures per capita tend to rise as density increases (Ladd and Yinger 1991; Ladd 1992; Ladd 1994; Holcombe and Williams 2008). Other works show the opposite conclusion (Carruthers and Ulfarsson 2003; Burchell and Mukherji 2003; Litman 2004; Carruthers and Ulfarsson 2008). Finally, other studies find little significance (Cox and Utt 2004). Following Bodkin and Conklin (1971), the relationship between population density and expenditures depends on the category of expenditure at stake.

In the same way, the literature has gone further to explore the relationship between population growth and public expenditures. Pettersson-Lidbom (2001), Hulten and Peterson (1984), and Rivers and Yates (1997) argue theoretically that population growth causes a rise in the requirement for municipal expenditures. The same relationship is empirically found by Farnham (1985). However, Ladd (1992), Cox and Utt (2004), Holcombe and Williams (2008) and Carruthers and Ulfarsson (2008) indicate that higher population growth is not associated with higher per capita total expenditures. Bodkin and Conklin (1971) find different impacts of population growth depending on the type of municipal expenditure.

Regarding other variables connected with urban development, Holcombe and Williams (2008) note that higher median house values are positively related to the level of government expenditures. Cox and Utt (2004) find a positive correlation of median house value and municipal spending. Oates (1969) supports the same relationship with municipal expenditures on public schools. Carruthers and Ulfarsson (2003) show mixed different results for median house value depending on the municipal expenditure at stake.

The literature has also considered LG size, urbanized area and housing age as factors that influence LG spending (Cox and Utt 2004; Carruthers and Ulfarsson 2003, 2008).

Politics

The political literature has shown that both partisan politics and political strength influence LG financial situation.

According to partisan politics theory, governments' ideology shapes policies and outcomes. It is commonly assumed that left-wing parties favor public spending increases while right-wing parties aim at budget reductions (Seitz 2000; Tellier 2006). According to Allers et al. (2001) left-wing political parties are believed to be more in favor of an active state and income redistribution than right-wing parties. Cusack (1997) defines this idea as the "partisan politics matters" (PPM) theory. Blais and Nadeau (1992) empirically prove this theory in a survey on 10 Canadian provinces. They find that right-wing parties spend 3% less in per capita terms, and their deficit is lower. Dickson and Yu (1997) and Tellier (2006),

using again Canadian provinces, find empirical support for the PPM thesis. Similarly, Allers et al. (2001) prove that the more left-wing the LG is, the higher the tax burden is, in a sample of Dutch LG.

Nevertheless, though most of the literature supports the partisan politics theory, empirical findings are not totally conclusive. For instance, Abizadeh and Gray (1993), Bosch and Suarez-Pandiello (1995), Galli and Rossi (2002), Seitz (2000), Benito and Bastida (2004) and Hagen and Vabo (2005) fail to prove it at the municipal level. Furthermore, Lago-Peñas and Lago-Peñas (2009) show in a recent study that incumbent's ideology is not relevant to explain deviations in Spanish LG deficit. All these works are connected with the so-called convergence school of thought: modern industrialised societies have become increasingly similar, facing the same kind of problems and thus using the same kind of solutions. Consequently, political, institutional and cultural differences do not matter much when it comes to explain variations in policy outputs (Skinner 1976; Thomas 1980).

However, so far, most of the literature has focused on the "partisan politics matters" thesis at the national level (Allers et al. 2001). This paper is meant to contribute to the sub-national PPM literature, addressing the claim of Imbeau et al. (2001) of further research. As stated above, differences in party composition of government, in general, matter in public policy in constitutional democracies. Nevertheless, we must bear in mind that Spanish municipal politicians are involved in solving direct and practical citizens' needs. Therefore, there is no place to redistribute income or implement other policies directly connected to the political orientation of parties, as we can find in national politics. Although most democracies face anti-incumbent voting in regional and local elections, local elections are about the provision of local services (Travers 2009). Accordingly, the partisan influence on Spanish LG is to some extent uncertain, and thus we expect a confirmation of the "convergence" school of thought.

With respect to political strength, there are two approaches in the theoretical debate over the influence of divided governments on the fiscal situation of public entities (Alt and Lowry 1994). On the one hand, Roubini and Sachs (1989a, b) defend that large coalition governments face higher deficits and spending levels than one-party incumbents governing with a majority. They suggest that coordination problems in divided governments can impose noteworthy costs. This theory is known in the literature as Roubini and Sachs' weak government hypothesis (RSH). On the other hand, Alesina and Rosenthal (1994) think that divided governments have a moderating influence on fiscal policy. Jones et al. (1997) investigate political strength from the point of view of budget volatility. They consider that dissensual politics lead to small budget changes. This is the opposite of what RSH predicts: weak governments increase budget expenditures as a result of their inability to resist interest groups' pressures. What the newer literature shows is that not only the median voter matters but so does the variance: the smaller the variance in voter preferences, the greater the political/legislative consensus, the stronger the government.

Control Variables

Demographics

The demographic features have to do with the "interest groups problem", i.e., groups of population that pressure politicians to meet their needs. The literature has used many demographic variables to account for these interest groups, among them rate of young population and rate of immigrants (Cox and Utt 2004; Hagen and Vabo 2005; Carruthers and Ulfarsson 2008). We include

two variables that capture the influence of interest groups on spending, namely, young population and immigration.

There is a dispute between young and old about spending decisions (Hoyt and Toma 1993). Poterba (1998) concludes that this relationship is complex from a theoretical point of view and that further research is needed on this issue. Poterba (1997) and Grob and Wolter (2007) show that indeed there is an impact on education. Borge and Rattso (1995) find that a higher proportion of young people leads to higher spending on education. These results support the fact that interest groups struggle for pieces of the public spending pie.

With regard to immigration, Grob and Wolter (2007) assert that its impact on spending is a complex issue. Alesina et al. (1999) find, for US local governments, that the higher the immigration, the higher the deficit and debt per capita. This finding agrees with political economy theories: heterogeneous societies value public goods less and are not concerned about fiscal discipline. They state that immigration, as an interest group, promotes an increase in targeted expenditure at the expense of public goods provision such as education (or infrastructure: Easterly and Levine 1997). They also prove that, in agreement with the log-rolling theory, total spending, health spending per capita and police spending per capita are positively related to immigration.

Other demographic variables may also have an impact on public expenditure. This way, Cox and Utt (2004) and Carruthers and Ulfarsson (2008) find a negative and significant relationship between the number of people per household and local spending.

Fiscal/Economics

According to the traditional theory on subnational government spending, the economic variables that are believed to impact local expenditures are real per capita personal income, per capita real intergovernmental revenue and per capita real unemployment compensation (Bails and Tieslau 2000).

Regarding municipal income level, it is another "classical variable" used by Fabricant (1952) to analyze LG financial situation. Furthermore, Wagner's Law shows that income explains government growth in the states, and even more, in the local authorities (Wagner 1958). Thus, Pettersson-Lidbom (2001) controls for this variable, since it is related to the LG fiscal capacity, as most of the revenues come from a proportional local income tax. We agree with this literature, and accordingly, we consider the economic level as a control variable in our model.

Ashworth et al. (2005) argue that the level of income will have a positive effect on public expenditures. Kiewiet and Szalaky (1996) and Dickson and Yu (1997) show LG spending is linked to increases in per capita income. In addition, Farnham (1985) states that the per capita income variable could pick up the influence of a positive income elasticity of demand for capital goods, in which case its impact on expenditures would be positive. The level of income (measured by disposable family income) causes a higher demand for infrastructures and other kinds of expenditures (Hulten and Peterson 1984). Oates (1969), Carruthers and Ulfarsson (2008), and Holcombe and Williams (2008) control for this variable since it is related to LG spending. The financing of the construction and maintenance of these capital goods has become a contentious issue because of the various demands placed on local government budgets (Najafi et al. 2007).

The literature has paid attention to the interaction between income and grants at the sub-national level. It is possible to set a theoretical prediction of the effect of grants on spending. The empirical evidence shows that spending is stimulated by much more than theory predicts. For unrestricted grants, the estimated effects are often closer to 100%. This result was dubbed the flypaper effect

by Okun (1981), since the money the government sends out “sticks where it hits” (Hines and Richard 1995).

Tovmo and Falch (2002) LG behavior models predict equal effects of private income and unconditional federal grants on LG expenditures. However, empirical studies find that the effect of grants is larger than the income effect. Dahlberg et al. (2008) find that grants are shifted to spending, but not to taxes. Strumpf (1998) proposes an index based on administrative expenditure which proxies for the level of voter information and helps explain the diversity of flypaper consumption. Lalvani (2002) finds that expenditures receive a greater stimulus from grants than would an increase in income. Levaggi and Zanola (2003) show that own resources is the least reactive item, since governments prefer to use grants especially if they are increasing private expenditure and deficit. Lago-Peñas (2008) tests asymmetries in the effects of increases and decreases in grants on total spending and he explains them with two factors: incumbent’s ideology and financial capacity of getting into debt.

The ratio of employment to population is hypothesised to influence LG spending per capita. The literature has found mixed results when analysing this influence. According to Carruthers and Ulfarsson (2008), this variable measures competition in the job market, therefore weak employment markets are expected to influence spending negatively. Holcombe and Williams (2008) also predict the same negative relationship. However, Bails and Tieslau (2000) find that the higher the level of unemployment compensation, the higher the level of per capita local spending. Considering a given type of spending, Marlow and Shiers (1999) hypothesize that unemployment is positively related to crime rates because in those states with relatively fewer jobs, there may be a greater incentive for criminal activity. This greater activity is connected with a higher expenditure on security per capita.

Some works find that the impact of unemployment depends on the type of spending. Foucault et al. (2008) find a positive and significant relationship between unemployment rate and current expenditures. However, this variable has a negative impact on investment, suggesting a substitution effect between current expenditures and investment expenditures in those LG that face unemployment.

Our model controls for the impact of taxes on spending, since a reduction in taxes generally leads to a reduction in LG spending (Gabe and Bell 2004). Similarly, Brown (1996) stated that the percentage of taxes out of total revenues is a good indicator for the LG financial independence. If this ratio is low, LG are unable to mobilize resources in their own jurisdictions to finance an increasing spending (De Mello 2001).

Econometric Procedure

Model

The characteristics of our dataset may lead to spatial effects, such as spatial autocorrelation or spatial dependence and spatial heterogeneity. The former appears when “LG *i*” spending depends on “LG *i*” characteristics and also on “LG *i*” neighbors’ spending: “everything is related to everything else, but near things are more related than distant things” (Tobler 1979). Since our sample is made up of spatially distributed data, neighbors’ influence (spillover effect) must be tackled. The latter exists when “LG *i*” features are influenced by its geographical situation. For example, in our sample, rich municipalities are located in the north east part of Spain (see Fig. 1).

In order to obtain accurate coefficients, it is important to control for both spatial autocorrelation and spatial heterogeneity through specific techniques (Paelinck and Klaasen 1979; Cliff and Ord 1972, 1973, 1981; Anselin 1980; Arbia 1989). The nonspatial econometrics can mostly control for the spatial heterogeneity. Our spatial model then controls for the spatial autocorrelation, since it cannot be solved by other techniques.

There are two primary types of spatial dependence: spatial lag (the dependent variables are correlated) and spatial error (the error terms across different spatial units are correlated). This spatial dependence may be caused by two reasons. First, data collection of observations associated with spatial units may reflect measurement error. This happens when the boundaries for which information is collected do not accurately reflect the nature of the underlying process generating the sample data. Second, the spatial dimension of a social or economic characteristic may be an important aspect of the phenomenon.

With spatial lag and spatial error in ordinary least squares regression, the hypothesis of independent observations and the assumption of uncorrelated error terms are violated. As a result, the estimates are biased and inefficient. In this way, spatial lag generates a transmission process: the behavior of a municipality leads to a similar behavior of neighboring municipalities. If this is the case, spatial error is indicative of omitted (spatially correlated) covariates that, if not controlled, would impair inference. Therefore, we develop two models that consider both forms of spatial dependence.

Accordingly, our models assume the neighbors’ influence on municipal spending. The reaction function of a LG may be written as (Brueckner and Saavedra 2001)

$$E_i = R(E_j, X_i) \quad (1)$$

where E_i = spending of LG *i*; E_j = LG *i* neighbors’ spending; and X_i = LG characteristics.

Starting from this function, we develop the first model which is known in the literature as a spatial autoregressive or spatial lag model

$$E = \rho WE + X\beta + u \quad (2)$$

where E = municipal spending per capita; WE (weight matrix) = spatial lag of variable E ; X = matrix of K exogenous variables; u = white noise, and finally ρ = autoregressive parameter that captures the intensity of interdependences among observations.

WE was built according to the queen contiguity rotation (variable *spatial lag*). This yields a binary contiguity matrix, where w_{ij} takes value “1” if observations i and j are adjacent, and value “0” otherwise.

If Eq. (1) is the correct model, then ignoring the spatial autocorrelation term means that a significant explanatory variable has been omitted. The consequence is that the estimates of β are biased and all statistical inferences are invalid.

We also consider that this model may suffer from endogeneity, since the spending of a LG may be influenced by their neighbors’ spending and vice versa. To control for this endogeneity, we use instrumental variables (IV) through a spatial two-stage least squares model (S2SLS).

To instrument WE , we take the exogenous variables (X) and their spatial lag (WX) as IV. Our choice of IV agrees with previous literature. Kelejian and Robinson (1992) took these IV for the first time, after proving that the most accurate IV when using contiguity matrix are the spatially lagged exogenous variables. Later, further literature has applied the same IV: Fingleton and López-Bazo (2003), Carruthers and Ulfarsson (2008).

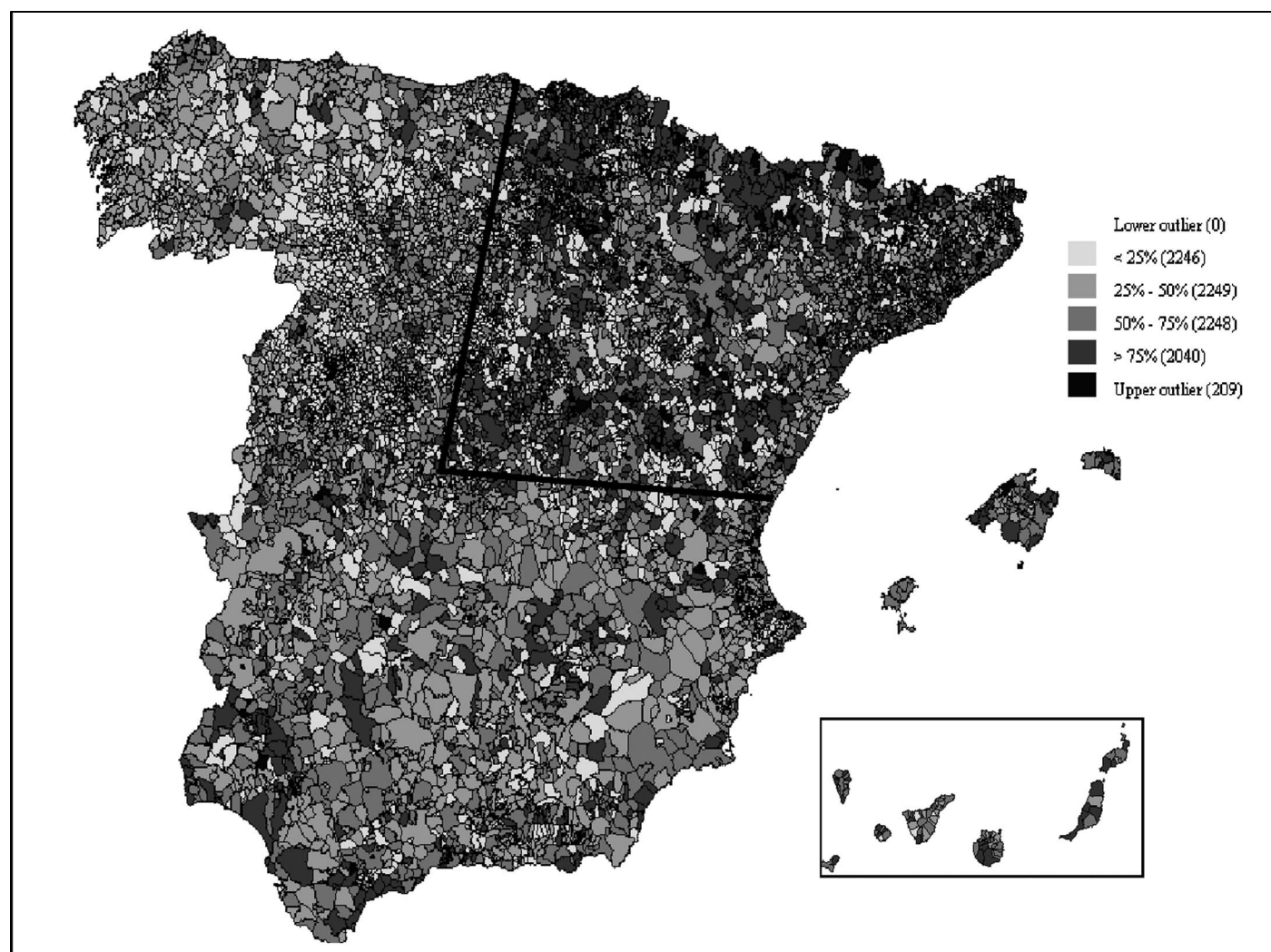


Fig. 1. Spatial distribution of total LG spending per capita in Spain

Furthermore, there may be an endogeneity problem with tax revenues and property tax revenues. High preferences for spending are translated to higher tax rates and thus to higher revenues. Moreover, if the property tax is the tax used in the margin to fund additional improvements in local services, higher preferences for public spending might also translate to a higher reliance on the property tax. Accordingly, we have run endogeneity tests and results suggest the non-existence of endogeneity.

The other type of spatial dependence is represented by a spatial error model. A spatial error model can be expressed as

$$Y = X\beta + \varepsilon \quad (3)$$

Where spatial autocorrelation is reflected in the following error term:

$$\varepsilon = \lambda W\varepsilon + \mu \quad (4)$$

where λ = spatial autoregressive coefficient and is assumed to lie between -1 and $+1$; and μ = n -element vector of error terms. The coefficient λ measures how neighboring observations affect the dependent variable, but the interpretation differs from that of a spatial lag model. In the spatial error model, LG spending is affected by a shock to spending in neighboring LG. In other words, a shock in neighboring LG spills over to a degree depending on the value of λ

through the error term. If the spatial autocorrelation represented by equation (4) is erroneously ignored, then similar to the spatial lag case, standard statistical inferences are invalid although the coefficients remain unbiased.

Sample

The sample is made up of all Spanish municipalities with more than 1,000 inhabitants (3,204 LG) for the year 2005. This sample represents nearly 40 percent of the population (8,112 LG). The reason for eliminating LG with less than 1,000 inhabitants is twofold. On the one hand, the reliability of the financial data is doubtful for small LG. On the other hand, some variables were only available for LG with more than 1,000 inhabitants. According to Pettersson-Lidbom (2001) and Ashworth et al. (2005), municipal datasets have two main advantages over cross-country datasets, namely, homogeneity and amplitude. First, heterogeneity (different legal structures and socioeconomic framework in cross-country samples) needs to be controlled (Blejer and Cheasty 1991; Tellier 2006). This feature is overcome in the municipal level within a country. Second, subnational datasets are considerably larger than cross-country samples. Finally, according to Poterba (1995), international comparisons of budget rules and public spending are problematic. The many nonbudgetary differences between countries make it

Table 1. Definition of Variables and Descriptive Statistics

Variable group	Variable label	Description	Calculation	Mean	Median	St. dev.	Min.	Max.
Dependent variables	<i>expendtot</i>	2005 total expenditure per capita in €	Taken from Spanish Ministry of Treasury	894.89	785.97	446.67	244.15	6,813.93
	<i>solidwastch</i>	2005 garbage expenditure per capita in €		38.10	26.55	40.66	0.00	320.53
	<i>waterh</i>	2005 water supply expenditure per capita in €		28.92	15.18	44.14	0.00	709.61
	<i>housingh</i>	2005 housing expenditure per capita in €		132.66	95.53	186.25	0.00	5,770.82
	<i>securityh</i>	2005 civil protection and public safety expenditure per capita in €		31.93	23.82	37.06	0.00	398.41
Urbanization/ geography	<i>educationh</i>	2005 education expenditure per capita in €		42.30	24.34	92.60	0.00	3,825.42
	<i>cultureh</i>	2005 culture expenditure per capita in €		59.13	43.02	67.34	0.00	935.90
	<i>popdens</i>	2005 Population density of LG	Population/Urbanized km ² . Population taken from Spanish National Statistics Institute. Urbanized km ² taken from Property Assessment Office	6,587.91	5,596.51	4,803.48	145.88	62,505.77
	<i>pg_01_05</i>	Rate of population growth between 2001 and 2005	$\frac{\sqrt[4]{2005 \text{ population}} - 1}{\sqrt[4]{2001 \text{ population}}}$	0.01	0.01	0.03	-0.08	0.25
	<i>sizeurbarea</i>	Size of the municipality in km ² Built area in 2005 (%)	Taken from Spanish National Statistics Institute Urbanized land area/total land area. Both taken from Property Assessment Office	97.50 4.12	56.41 1.38	132.42 6.98	0.39 0.04	1,750.33 63.90
Politics	<i>housingvalue</i>	2005 median housing value of the municipality in thousand €	Taken from Property Assessment Office	28.77	23.32	22.68	3.17	360.11
	<i>housingage</i>	2001 median housing age of the municipality in years	Taken from National Statistics Institute	20.81	20.09	6.84	2.26	55.95
	<i>ideology</i>	LG political ideology	Political ideology of the municipal council (0-left; 1-right). Taken from Ministry of Internal Affairs	0.47	0.00	0.50	0.00	1.00
	<i>polstrength</i>	Herfindahl index measuring LG political strength	Continuous variable ranging from 0 (minimum political strength) to 1 (maximum political strength)	0.46	0.44	0.13	0.20	1.00
	<i>localpation</i>	Local parties influence National party effect	Local parties' seats/total LG seats Takes value 1 if national and municipal party are the same; 0 otherwise	0.15 0.42	0.00 0.00	0.22 0.49	0.00 0.00	1.00 1.00
Demo-graphics	<i>region</i>	Regional party effect	Takes value 1 if regional and municipal party are the same; 0 otherwise	0.54	1.00	0.50	0.00	1.00
	<i>immigrants</i>	Percent of immigrants of the municipality in 2005	Taken from Spanish National Statistics Institute	6.31	3.95	7.30	0.04	69.55
	<i>popless5</i>	Percent of population less than 5 years old of the municipality in 2005		4.32	4.39	1.54	0.22	12.99
	<i>houseize</i>	Average household size of the municipality in 2005		2.84	2.83	.263	1.47	4.26
	<i>transfersh</i>	Regional and central transfers per capita in €	Taken from Spanish Ministry of Treasury	411.62	351.60	242.66	108.46	4,642.39
Fiscal/ economics	<i>taxrevenueh</i>	Tax revenue per capita in €		280.15	231.25	231.87	22.27	5,019.76
	<i>taxproperty</i>	Rate of tax revenue coming from property taxes	Property taxes/Total taxes	0.42	0.43	0.12	0.04	0.88
	<i>incomeh</i>	2005 income per capita in €	Taken from Lawrence R. Klein Institute	12,006.29	11,449.07	2,679.81	6,442.42	29,328.24
	<i>%unemploy</i>	2005 rate of unemployed population of the municipality (%)	Taken from Anuario Económico "La Caixa"	4.08	3.70	1.87	0.60	13.70

difficult to attribute differences in spending patterns only to fiscal institutions. Among other works having used sub-national data we point out: Borge (1995), Alt and Lowry (1994), Poterba (1994), Pettersson-Lidbom (2001), Ashworth et al. (2005), and Hagen and Vabo (2005). Regarding the problems of working with subnational data, we can mention that international comparative analysis is not possible, preventing us from assessing the differences in countries' municipal spending.

We take year 2005 to avoid an electoral bias. Municipal elections took place in Spain in 2003 and 2007. Therefore, 2005 is the most recent year that is expected to be free from the "municipal electoral cycle" (the next one would be 2009). This cycle has been investigated extensively by the literature, and stems from politicians' discretionary power on public budget to increase re-election chances. Thus, pre-election year (2006) is influenced by the incumbents' strategy of spending increase in order to be re-elected, whereas the fiscal adjustment occurs in the post-election year (2004).

Variables

The variables in our model have been considered in agreement with the theoretical and empirical underpinnings (Table 1 depicts descriptive statistics). Expenditure variables have been calculated on a per capita basis, to make possible the comparison among LG. All variables refer to 2005, except for *housing*, which was not available for 2005 and was drawn from 2001 National Statistics Institute Census.

We analyse seven expenditure dependent variables (all of them in per capita terms): total expenditures (*expendtoth*), disposal of garbage (*solidwasteh*), water supply (*waterh*), public housing (*housingh*), public safety (*securityh*), public education (*educationh*) and culture (*cultureh*).

The literature on LG spending has studied a mix of explanatory variables. We have organized them into two categories: urbanization/geography and politics. Our model is completed with control variables grouped in two areas: demographics and fiscal/economics. Other works using similar categories are, among others, Bodkin and Conklin (1971), Holcombe and Williams (2008) and Carruthers and Ulfarsson (2008).

The first category is made up of different geographical features of the municipality: population density, size, urbanized area, housing value and housing age.

The second category is made up of politics characteristics such as political ideology and strength, local parties' influence and national and regional political effects. Regarding the two latter variables, Solé-Ollé and Sorribas-Navarro (2008) find that Spanish LGs aligned with upper-tier grantor governments (i.e., controlled by the same party) receive 40 percent more grants than unaligned ones. Accordingly, we include two dummy variables (*nation* and *region*) to take into account whether the LG is aligned with upper-tier governments (national and regional, respectively), thus

controlling for national/regional party effects in the regression. We have included the control variable *localp* (local parties) to account for the influence of these parties on the LG spending, but according to Allers et al. (2001), we have taken their political ideology as "neutral".

In respect of political strength, as suggested by the literature (Rattsø and Tovmo 2002; Borge 2005; Hagen and Vabo 2005), we calculate a Herfindahl index for each municipal council (variable *polstrength*). This approach allows us to test whether the strength of both bodies has an impact on LG fiscal situation

$$\frac{\sum_{i=1}^n S_i^2}{S^2}$$

where S = total number of seats (aldermen) in the municipal council; S_i = number of seats of party i in the municipal council. Fig. 1 shows the spatial distribution of total LG spending in Spain. We note that north east part of Spain (mainly Madrid, Navarre, Basque Country, and Catalonia) show higher levels of spending per capita (dark grey and black color in the map). Therefore, controlling spatial econometric issues could improve the estimations of LG spending in Spain. That is the reason why our models control for the spillover effect of neighboring LG on the LG at stake.

Results

Table 2 depicts the spatial tests applied. The first statistic to determine spatial dependence is Moran's I , since it is a spatial autocorrelation measure. Besides, the null hypothesis of the SARMA test (Anselin 1988) is the total absence of spatial effects in the model, both in the main equation and in the error term. On the one hand, Table 2 reveals that the Moran's I statistic is highly significant in all the regressions, showing strong spatial autocorrelation of the residuals. On the other hand, SARMA test rejects the null hypothesis of no-autocorrelation.

Furthermore, the simple statistics Lagrange multiplier (lag) and Lagrange multiplier (error) are significant, which also indicates presence of spatial dependence. The robust tests (robust LM-lag and robust LM-error) indicate what type of spatial dependence is present in the data. The robust measure for error remains significant. However, the robust lag test significance decreases in all cases, even becoming insignificant in three regressions (*waterh*, *educationh*, *cultureh*). This means that the error dependence lessens when lagged dependent variable is present.

Regarding the error tests, the pattern is similar. Robust test (robust LM error) presents clearly lower significance than the simple test. This feature also indicates that, to some extent, the error model corrects the spatial autocorrelation.

In sum, it appears that controlling for spatial lag and spatial error improves the specification of the ordinary least squares model.

Table 2. Diagnostics for Spatial Dependence

Test (z-values)	<i>expendtoth</i>	<i>solidwasteh</i>	<i>waterh</i>	<i>housingh</i>	<i>securityh</i>	<i>educationh</i>	<i>cultureh</i>
Moran's I (error)	5.599 ^c	17.701 ^c	14.693 ^c	5.698 ^c	17.259 ^c	4.453 ^c	4.012 ^c
Lagrange multiplier (lag)	13.380 ^c	302.410 ^c	179.688 ^c	32.828 ^c	344.103 ^c	18.556 ^c	13.320 ^c
Robust LM (lag)	3.977 ^b	45.452 ^c	3.638 ^a	5.547 ^b	79.115 ^c	0.963	1.237
Lagrange multiplier (error)	30.110 ^c	308.380 ^c	211.987 ^c	31.196 ^c	293.098 ^c	18.870 ^c	15.241 ^c
Robust LM (error)	20.707 ^c	51.421 ^c	35.946 ^c	3.914 ^b	28.111 ^c	1.276	3.158 ^a
Lagrange multiplier	34.087 ^c	353.831 ^c	215.625 ^c	36.742 ^c	372.214 ^c	19.833 ^c	16.478 ^c

^a1% significance.

^b5% significance.

^c10% significance.

Table 3. S2SLS, Spatial Lag, and Spatial Error Models

Variable category	Dependent variable: <i>expendloth</i>			Dependent variable: <i>solidwasteh</i>		
	S2SLS	Spatial lag	Spatial error	S2SLS	Spatial lag	Spatial error
Urbanization/geography	Spatial lag: WE					
	Lambda					
Urbanization/geography		0.007 ^b (2.197)	0.021 ^c (3.824)	0.052 ^c (7.918)	0.178 ^c (15.176)	0.223 ^c (14.726)
	<i>popdens</i>	−0.005 ^c (−4.294)	−0.005 ^c (−7.273)	0.000 ^c (3.277)	0.000 ^c (5.740)	0.001 ^c (6.316)
	<i>pg_01_05</i>	1,899,698 ^c (6.204)	1,570,809 ^c (9.948)	−73,935 ^c (−1.870)	−185,008 ^c (−8.952)	−176,655 ^c (−8.188)
	<i>size</i>	−0.045 (−0.932)	0.022 (1.081)	0.010 ^b (2.030)	0.018 ^c (6.792)	0.018 ^c (6.465)
	<i>urbarea</i>	0.570 (0.645)	0.159 (0.314)	0.719 ^c (6.374)	0.770 ^c (11.482)	0.942 ^c (13.440)
	<i>housevalue</i>	0.924 ^c (3.019)	0.216 (1.429)	0.058 (1.440)	0.103 ^c (5.212)	0.124 ^c (6.028)
Politics	<i>houseage</i>	−4.211 ^c (−3.382)	−4.507 ^c (−7.668)	−0.432 ^c (−2.688)	−0.702 ^c (−9.118)	−0.798 ^c (−9.979)
	<i>ideology</i>	7.488 (0.402)	−5.085 (−0.519)	−6.012 (−0.613)	−2.111 ^a (−1.648)	−2.284 ^b (−1.775)
	<i>polstrength</i>	286,101 ^c (5.916)	300,906 ^c (13.599)	297,526 ^c (13.372)	−7,199 ^b (−2.485)	−6,327 ^b (−2.156)
	<i>localp</i>	52,037 ^a (1.786)	26,191 ^a (1.958)	28,262 ^b (2.085)	15,732 ^c (8.984)	14,401 ^c (7.946)
	<i>nation</i>	−11,410 (−0.584)	−3,991 (−0.387)	−5,660 (−0.547)	−6,252 ^c (−4.636)	−6,067 ^c (−4.459)
	<i>region</i>	1,852 (0.164)	−4,997 (−0.829)	−3,972 (−0.657)	3,858 ^c (4.895)	3,778 ^c (4.753)
Demographics	<i>immigrants</i>	0.050 (0.059)	−0.994 ^b (−2.201)	−1,116 ^b (−2.394)	0.690 ^c (11.642)	0.704 ^c (10.908)
	<i>popless5</i>	3,241 (0.562)	−2,930 (−0.981)	−2,091 (−0.695)	0.549 (1.408)	0.915 ^b (2.285)
	<i>houseize</i>	−42,707 ^a (−1.841)	−4,516 (−0.469)	−2,515 (−0.257)	−4,846 ^c (−3.848)	−4,912 ^c (−3.719)
	<i>transfersh</i>	0.935 ^c	1,034 ^c (95.586)	1,037 ^c (95.711)	0.012 ^c (3.786)	0.011 ^c (7.857)
Fiscal/economics	<i>taxrevenueh</i>	0.923 ^c (32.024)	1,045 ^c (71.832)	1,042 ^c (71.149)	0.044 ^c (11.779)	0.040 ^c (20.790)
	<i>taxproperty</i>	75,224 ^a (1.811)	226,586 ^c (14.868)	225,397 ^c (14.719)	27,701 ^c (5.162)	15,222 ^c (7.445)
	<i>incomeh</i>	0.003 (0.939)	0.008 ^c (6.991)	0.009 ^c (7.046)	0.002 ^c (4.412)	0.002 ^c (12.987)
	<i>%unemploy</i>	−7,850 ^b (−2.430)	−9,354 ^c (−5.600)	−9,279 ^c (−5.412)	1,170 ^c (2.816)	0.837 ^c (3.567)
		N	N	N	N	N
	2,599	2,599	2,599	2,556	2,556	2,556
	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²	Adj. R ²
	0.637	0.904	0.904	0.361	0.549	0.551

Note: T-values and Z-values in parentheses.

^a5% significance.^b1% significance.^c10% significance.

Table 4. S2SLS, Spatial Lag, and Spatial Error Models

Variable category	Dependent variable: <i>waterh</i>			Dependent variable: <i>housingh</i>		
	S2SLS	Spatial lag	Spatial error	S2SLS	Spatial lag	Spatial error
Urbanization/geography	Spatial lag: <i>WE</i>					
	Lambda					
Politics						
Demographics						
Fiscal/economics						

Note: T-values and Z-values in parentheses.

^a1% significance.^b10% significance.^c5% significance.

Table 5. S2SLS, Spatial Lag, and Spatial Error Models

Variable category	Dependent variable: <i>securityh</i>			Dependent variable: <i>educationh</i>		
	S2SLS	Spatial lag	Spatial error	S2SLS	Spatial lag	Spatial error
Urbanization/geography	Spatial lag: <i>WE</i>					
	Lambda					
		0.039 ^c (6.226)	0.189 ^c (16.216)	0.058 ^c (5.379)	0.080 ^c (5.397)	0.092 ^c (5.540)
	<i>popdens</i>	0.001 ^c (5.394)	0.001 ^c (9.401)	0.000 (−1.207)	−0.000 (−0.424)	−0.000 (−0.248)
	<i>pg_01_05</i>	−189.462 ^c (−5.234)	−240.739 ^c (−13.115)	44.510 (0.496)	17.406 (0.349)	11.914 (0.234)
Politics	<i>size</i>	0.026 ^c (5.493)	0.030 ^c (12.696)	−0.024 ^c (−2.849)	−0.009 (−1.481)	−0.012 ^a (−1.852)
	<i>urbarea</i>	0.462 ^c (4.472)	0.438 ^c (7.392)	0.034 (0.184)	0.096 (0.601)	0.109 (0.665)
	<i>housvalue</i>	0.087 ^b (2.371)	0.034 ^c (1.942)	0.189 ^c (2.792)	0.177 ^c (3.717)	0.169 ^c (3.476)
	<i>housage</i>	−0.858 ^c (−5.861)	−1.184 ^c (−17.308)	−0.101 (−0.383)	−0.740 ^c (−3.972)	−0.876 ^c (−4.636)
	<i>ideology</i>	−2.145 (−0.975)	0.228 (0.200)	−1.347 (−0.339)	−16.664 ^c (−5.388)	−15.668 ^c (−5.047)
Demographics	<i>polstrength</i>	−15.491 ^c (−2.708)	−22.706 ^c (−8.829)	−5.200 (−0.505)	18.475 ^c (2.642)	17.172 ^c (2.438)
	<i>localp</i>	7.984 ^c (2.348)	8.111 ^c (5.218)	3.901 (0.636)	13.775 ^c (3.258)	13.178 ^c (3.068)
	<i>nation</i>	−4.302 ^a (−1.747)	−2.216 ^a (−1.851)	−5.488 (−1.320)	−16.119 ^c (−4.951)	−15.469 ^c (−4.725)
	<i>region</i>	4.894 ^c (3.676)	4.874 ^c (6.966)	−0.618 (−0.257)	−4.395 ^b (−2.309)	−4.095 ^b (−2.140)
	<i>immigrants</i>	0.881 ^c (8.601)	0.652 ^c (12.381)	0.020 (0.109)	−0.055 (−0.388)	−0.052 (−0.354)
Fiscal/economics	<i>popless5</i>	1.270 ^a (1.827)	1.129 ^c (3.251)	4.705 ^c (3.736)	2.463 ^c (2.616)	2.447 ^b (2.569)
	<i>houseize</i>	3.644 (1.350)	−0.026 (−0.023)	−1.440 (−0.296)	−5.735 ^c (−1.886)	−5.402 ^a (−1.740)
	<i>transfersh</i>	0.000 (−0.104)	0.005 ^c (4.423)	0.059 ^c (11.568)	0.081 ^c (23.595)	0.083 ^c (24.054)
	<i>taxrevenueh</i>	0.033 ^c (9.888)	0.035 ^c (20.461)	0.014 ^b (2.268)	0.026 ^c (5.620)	0.026 ^c (5.500)
	<i>taxproperty</i>	17.446 ^c (3.557)	10.593 ^c (6.022)	−3.858 (−0.436)	−12.359 ^c (−2.576)	−14.660 ^c (−3.022)
	<i>incomeh</i>	0.001 ^c (3.017)	0.002 ^c (12.924)	0.001 ^b (2.424)	0.002 ^c (6.589)	0.003 ^c (7.045)
	<i>%unemploy</i>	1.847 ^c (4.852)	1.571 ^c (8.088)	−0.802 (−1.171)	−0.389 (−0.739)	−0.256 (−0.469)
		N	N	N	N	N
		2,556	2,556	2,556	2,556	2,556
		Adj. <i>R</i> ²	Adj. <i>R</i> ²	Adj. <i>R</i> ²	Adj. <i>R</i> ²	Adj. <i>R</i> ²
		0.340	0.536	0.156	0.229	0.229

Note: T-values and Z-values in parentheses.

^a1% significance.^b10% significance.^c5% significance.

Table 6. S2SLS, Spatial Lag, and Spatial Error Models

Variable category		Dependent variable: <i>cultureh</i>		
		S2SLS	Spatial lag	Spatial error
Spatial lag: <i>WE</i>		0.015 (1.603)	0.041 ^c (3.262)	
Lambda				0.055 ^c (3.258)
Urbanization/geography	<i>popdens</i>	0.000 ^a (−1.879)	−0.001 ^c (−3.597)	−0.001 (−3.674)
	<i>pg_01_05</i>	177.186 ^c (2.703)	38.566 (1.019)	38.021 (0.993)
	<i>size</i>	−0.006 (−0.689)	−0.006 (−1.269)	−0.007 (−1.436)
	<i>urbarea</i>	−0.281 (−1.501)	−0.411 ^c (−3.404)	−0.391 ^c (−3.192)
	<i>housvalue</i>	0.152 ^b (2.329)	−0.100 ^c (−2.768)	−0.105 ^c (−2.863)
	<i>housage</i>	0.140 (0.530)	−0.681 ^c (−4.834)	−0.705 ^c (−4.953)
Politics	<i>ideology</i>	1.419 (0.356)	−4.529 ^a (−1.931)	−4.561 ^a (−1.940)
	<i>polstrength</i>	12.104 (1.171)	23.387 ^c (4.222)	23.158 ^c (4.160)
	<i>localp</i>	0.312 (0.050)	12.724 ^c (3.968)	12.994 ^c (4.014)
	<i>nation</i>	0.134 (0.032)	−6.037 ^b (−2.445)	−6.185 ^b (−2.496)
	<i>region</i>	−1.211 (−0.502)	0.731 (0.506)	0.912 (0.630)
Demographics	<i>immigrants</i>	0.587 ^c (3.204)	0.122 (1.125)	0.109 (0.982)
	<i>popless5</i>	0.471 (0.379)	−0.365 (−0.511)	−0.263 (−0.367)
	<i>housize</i>	0.809 (0.165)	−5.202 ^c (−2.256)	−4.942 ^b (−2.117)
Fiscal/economics	<i>transfersh</i>	0.056 ^c (10.976)	0.067 ^c (25.750)	0.067 ^c (25.753)
	<i>taxrevenueh</i>	0.037 ^c (6.067)	0.067 ^c (19.084)	0.067 ^c (19.005)
	<i>taxproperty</i>	−7.051 (−0.794)	−13.441 ^c (−3.682)	−14.516 ^c (−3.969)
	<i>incomeh</i>	0.003 ^c (4.764)	0.004 ^c (14.752)	0.004 ^c (14.860)
	<i>%unemploy</i>	−0.684 (−0.989)	−1.022 ^b (−2.557)	−1.034 ^b (−2.537)
	N	N	N	N
		2,556	2,556	2,556
		Adj. <i>R</i> ²	Adj. <i>R</i> ²	Adj. <i>R</i> ²
		0.159	0.460	0.460

Note: T-values and Z-values in parentheses.

^a1% significance.

^b10% significance.

^c5% significance.

Therefore, for each regression we present the three aforementioned models. As an additional indicator of the specification enhancement, adjusted *R*² slightly improve when spatial problems are controlled.

Tables 3–6 present the coefficients of the regressions. All equations include an intercept, which is not shown for the sake of simplicity.

Discussion

The main interest of this paper is to analyze the determinants of municipal expenditures in Spain. These factors are organized into two categories: urbanization/geography and politics, with control variables grouped in two areas: demographics and fiscal/economy. Our sample includes all of the Spanish LG with more than 1,000 inhabitants, which makes our results highly relevant. As explained above, our models control spatial autocorrelation (S2SLS, spatial lag, and spatial error). These models assume the existence of spatial dependence, i.e., the spending of a municipality is influenced by its neighboring municipalities' spending. The regressions show that the coefficients of spatial lag and lambda are positive and significant in all regressions.

Urbanization/Geography

The variable *popdens* shows a clear and significant impact on total expenditures per capita (*expendtoth*): the higher the population density, the lower the total expenditures per capita. This result confirms the second theoretical assumption of Bails and Tieslau (2000):

population density leads to economies of scale in the provision of LG services. If we relate our findings to Holcombe and Williams (2008), we reach to the opposite conclusion, i.e., higher population density in Spanish municipalities is connected with lower expenditures per capita. However, the results of Holcombe and Williams apply mainly to larger cities in their sample. As Lee and Chan (2008) suggest, governments should consider adopting compact design in their urban developments.

This evidence also confirms previous literature, such as Carruthers and Ulfarsson (2003, 2008), Burchell and Mukherji (2003), Cox and Utt (2004), and Litman (2004). However, the impact of population density on the majority of expenditures is small (coefficients around zero).

Most of the previous theoretical and empirical works indicate a link between population growth and expenditures. The impact of population growth (*pg_01_05*) is positive and significant for total expenditures, water, housing and culture. Our results are in line with Hulten and Peterson (1984), Farnham (1985), Rivers and Yates (1997), and Pettersson-Lidbom (2001).

We do not find a clear pattern regarding the size of the municipality. With respect to *urbarea*, the results are also mixed, showing a positive impact for *solidwasteh*, *housing* and *securityh*. On the contrary, we find a negative impact for *waterh* and *cultureh*.

The impact of *housvalue* on the majority of expenditures is positive (Holcombe and Williams 2008; Cox and Utt 2004). It is noteworthy that our results in education confirm the findings of Oates (1969) on public schools.

There is a negative relationship between the median house age (*housage*) and all the expenditure categories except *waterh*.

Political Factors

The variable *ideology* does not present a clear behavior. Both the no-significance in *expendtoth* regressions and the mixed sign of the coefficients in the remaining regressions prevent us from finding out the impact of ideology on Spanish LG spending. This result agrees with previous literature: Abizadeh and Gray (1993), Benito and Bastida (2004) and Hagen and Vabo (2005). This literature is part of the “convergence” school of thought at the national level. However, the reason of this “ideology does not matter” conclusion is different: LG solve people’s needs (Travers 2009), with small influence of ideology (“no divergence”), whereas national governments have “converged” in their policies during the last decades.

If we pay attention to political strength (*polstrength*), we can say that it does affect LG spending: the lower the political fragmentation, the higher the spending. Consequently, municipal incumbents in Spain are using their majority power to implement their policies by means of more expenses and increasing their share in the local economy. This result is in line with Alesina and Rosenthal (1994) and Jones et al. (1997).

In sum, ideology does not influence spending. Nevertheless, strong political governments do spend more. Therefore, it is not that the incumbents belong to left or right, but it is their power (strong majority) which allows them to carry out their policies by spending more in per capita terms. The policy issue at stake here is whether political majorities in Spanish LG should be tempered. Municipal elections in Spain do not follow the “winner takes all” rule, but de facto, majorities make incumbents spend without having to negotiate with opposition parties. Some legislation, from our point of view, should be passed to enhance a Montesquieu approach. There is not a legislative control in LG, but if surveillance

mechanisms are enhanced, more effective control on spending could be made by the opposition.

Local parties play a small role in the LG financial situation (variable *localp*). Their importance in the municipal governments is small, with a mean of 15 percent of seats. They have a positive impact on total expenditures but with a low significance. However, local parties indeed make an influence on security (*securityh*) and solid waste (*solidwasteh*).

If we pay attention to variables *region* and *nation*, they do not show a clear pattern. Since previous literature had found that Spanish LG aligned with upper-tier governments received more grants than unaligned ones, we have included two dummy variables (*nation* and *region*) to find out whether this circumstance influenced spending. We can conclude that, unlike transfers, spending is not influenced by the alignment with upper governments.

Control Variables

The estimation results for the percentage of immigrants do not show a clear pattern. According to Grob and Wolter (2007), we prove that the relationship between immigration and LG spending is a complex issue, since it depends on each type of local expenditure (Carruthers and Ulfarsson 2008). Thus, immigration only has a positive impact on *solidwasteh* and *securityh*.

The results confirm three findings of Alesina et al. (1999). Firstly, ethnic fragmentation is positively correlated with police spending (*securityh*). Secondly, immigration, as an interest group, promotes an increase in targeted expenditure (in our case, *solidwasteh* and *securityh*) at the expense of public goods provision such as education (*educationh*). Similarly, Easterly and Levine (1997) showed that according to interest group theories, polarization leads

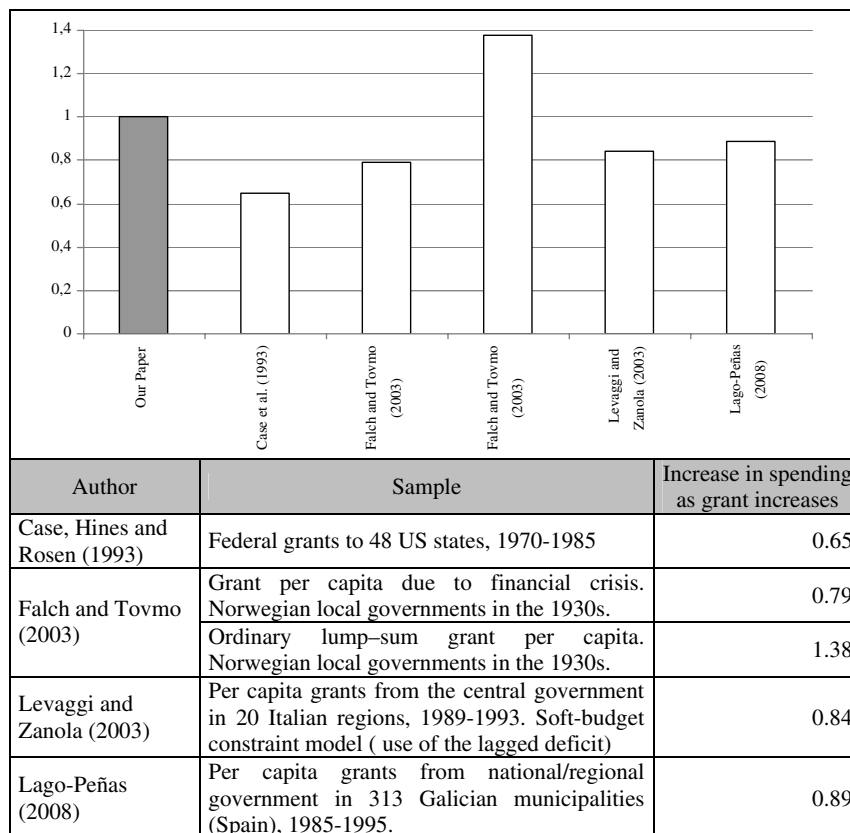


Fig. 2. This paper compared to the literature on flypaper effect

to rent-seeking behavior and reduces the consensus for public goods. Thirdly, in agreement with the log-rolling theory, police spending per capita (*securityh*) is positively related to immigration.

The share of population under five (*popless5*) has no impact on total spending. However, we find a relationship between young population and LG education spending (*educationh*), thus confirming Borge and Rattso (1995), Poterba (1997), Grob and Wolter (2007) and Carruthers and Ulfarsson (2008). This result was expected, since municipal competences on education are limited to kindergarten and child education.

As expected, the level of income is connected with higher expenditure per capita in all the categories. Our results are in line with the literature (Kiewiet and Szalaky 1996; Dickson and Yu 1997; Ashworth et al. 2005).

Regarding the unemployment rate (*unemploy*), our results also agree with the literature, which has found mixed results. On the one hand, unemployment influences negatively total spending per capita (*expendoth*), which confirms Holcombe and Williams (2008) and Carruthers and Ulfarsson (2008). On the other hand, it has a significant positive impact on security spending (*securityh*), in agreement with Marlow and Shiers (1999).

The positive effect of grants (*transfersh*) on spending is conclusive, not only in total spending, but also in the rest of spending categories. The average size of grants effect on total spending considering our three models is 1.002. This means that an increase of transfer of 1 Euro per capita leads to an increase of 1 Euro in per capita spending. This impact differs from the effect of income: 1 Euro increase in per capita income only leads to 0.007 Euro increase in total spending.

Unlike Tovmo and Falch (2002), who found equal effects of income and grants on LG expenditures, our results show that the impact of grants is much larger than that of the income (Lalvani 2002). This shows that Spanish LG spend grant money as if it were "free" money. So we prove a very strong flypaper effect.

If we compare with previous *flypaper* research, our coefficient falls within the range established by the literature (see Fig. 2). However, we can highlight that our coefficient is higher than that revealed the most recent literature.

Our model measures the effect of taxes on spending. On the one hand, the coefficients of variable *taxrevenueh* meet accurately the theoretical and empirical assumptions (Gabe and Bell 2004; Brown 1996; De Mello 2001). The impact of taxes on spending is positive in all the regressions, i.e., LG are using the tax receipts to increase spending rather than increasing municipal savings or reducing the level of indebtedness. On the other hand, the weight of property tax in LG total revenues (*taxproperty*) influences positively total spending, but it does not show a clear pattern in some spending categories.

Conclusions

The aim of this paper is to evaluate the determinants of Spanish municipal spending by estimating three models (S2SLS, spatial lag and spatial error) on all municipalities with more than 1,000 inhabitants (3,204 municipalities) in 2005. We study total municipal spending together with six categories of spending. Our results show that there are some interactions among neighboring municipalities in all LG expenditures. These relationships are positive: the spending of a municipality is influenced by its neighboring municipalities' spending.

We find a mixed influence of economic, geographic, politic and demographic factors on municipal expenditures per capita. We confirm the importance of geographic and political factors.

Concerning geographical factors, our data show that population density has a negative impact on municipal expenditures per capita while population growth has a positive impact. The negative impact of population density on per capita total expenditures has policy relevance because it indicates that policies aiming at increase population density will reduce per capita government expenditures, both in small and larger cities. Therefore, our results support the idea that sprawling development increases government costs relative to more compact urban development.

With respect to political factors, we find that political ideology does not make a clear influence on LG expenditures. Nevertheless, the level of municipalities' fragmentation does affect their finance situation: the lower the political fragmentation, the higher the spending.

LG in Spain are responding to their citizens' demands of increased range of public services, however there has not been a parallel transfer of financial resources so that LG are able to fund their increasing levels of spending. If we take France, for example, its level of decentralization is similar to Spain. However, legislation was passed so that French LG were guaranteed larger budgets and more staff to achieve their duties. Furthermore, our neighbor country passed further legislation to set LG responsibilities clear. In terms of planning and setting the scheme of duties, similar legislation should be issued in Spain, because LG, being the closest administration to citizens, are obliged to provide immediate services that sometimes are not appropriately funded.

Acknowledgments

The authors acknowledge the financial support of the Spanish Ministry of Economy and Competitiveness (ECO2010-17463 and ECO2010-20522).

References

- Abizadeh, S., and Gray, J. (1993). "Provincial government spending in Canada: An empirical analysis." *Int. Rev. Appl. Econ.*, 7(1), 69–90.
- Alesina, A., Baqir, R., and Easterly, W. (1999). "Public goods and ethnic divisions." *Q. J. Econ.*, 114(4), 1243–1284.
- Alesina, A., and Rosenthal, H. (1994). *Partisan politics, divided governments, and the economy*, Cambridge University Press, New York.
- Allers, M., De Haan, J., and Sterks, C. (2001). "Partisan influence on the local tax burden in the Netherlands." *Public Choice*, 106(3–4), 351–363.
- Alt, J. E., and Lowry, R. C. (1994). "Divided government, fiscal institutions, and budget deficits: Evidence from the states." *Am. Polit. Sci. Rev.*, 88(4), 811–828.
- Anselin, L. (1980). *Estimation methods for spatial autoregressive structures*, Regional Science Dissertation and Monograph Series, Ithaca, New York.
- Anselin, L. (1988). *Spatial Econometrics: Methods and Models*, Kluwer Academic Publishers, Boston.
- Arbia, G. (1989). *Spatial data configuration in statistical analysis of regional economics and related problems*, Kluwer, Dordrecht, The Netherlands.
- Ashworth, J., Geys, B., and Heyndels, B. (2005). "Government weakness and local public debt development in Flemish municipalities." *Int. Tax Publ. Finance*, 12(4), 395–422.
- Bails, D., and Tieslau, M. A. (2000). "The impact of fiscal constitutions on state and local expenditures." *Cato J.*, 20(2), 255–277.
- Benito, B., and Bastida, F. (2004). "The determinants of the municipal debt policy in Spain." *J. Publ. Budgeting Account Finance Manage.*, 16(4), 525–558.
- Blais, A., and Nadeau, R. (1992). "The electoral budget cycle." *Public Choice*, 74(4), 389–403.

- Blejer, M. I., and Cheasty, A. (1991). "The measurement of fiscal deficits: Analytical and methodological issues." *J. Econ. Lit.*, 29(4), 1644–1678.
- Brown, K. W. (1996). "Trend in key ratios using the GFOA financial indicators databases 1989–1993." *Government Finance Review*, 30–34.
- Bodkin, R. G., and Conklin, D. W. (1971). "Scale and other determinants of municipal government expenditures in Ontario: A quantitative analysis." *Int. Econ. Rev.*, 12(3), 465–481.
- Borge, L. E. (1995). "Economic and political determinants of fee income in Norwegian local governments." *Public Choice*, 83(3–4), 353–373.
- Borge, L. E. (2005). "Strong politicians, small deficits: Evidence from Norwegian local governments." *Eur. J. Polit. Econ.*, 21(2), 325–344.
- Borge, L. E., and Rattso, J. (1995). "Demographic shift, relative costs and the allocation of local public consumption in Norway." *Reg. Sci. Urban Econ.*, 25(6), 705–726.
- Bosch, N., and Suarez-Pandiello, J. (1995). "Seven hypotheses about public choice and local spending: A test for Spanish municipalities." *Public Finance*, 50(1), 36–50.
- Brueckner, J. I., and Saavedra, L. A. (2001). "Do local governments engage in strategic property-tax competition?" *Natl. Tax J.*, 54(2), 203–229.
- Burchell, R., and Mukherji, S. (2003). "Conventional development versus managed growth: The costs of sprawl." *Am. J. Public Health*, 93(9), 1534–1540.
- Carruthers, J. I., and Ulfarsson, G. F. (2003). "Urban sprawl and the cost of public services." *Environ. Plann. B: Plann. Des.*, 30(4), 503–522.
- Carruthers, J. I., and Ulfarsson, G. F. (2008). "Does smart growth matter to public finance?" *Urban Stud.*, 45(9), 1791–1820.
- Case, A. C., Hines, J. R., and Rosen, H. S. (1993). "Budget spillovers and fiscal policy interdependence: Evidence from the states." *J. Public Econ.*, 52(3), 285–307.
- Cliff, A., and Ord, J. (1972). "Testing for spatial autocorrelation among regression residuals." *Geogr. Anal.*, 4(3), 267–284.
- Cliff, A., and Ord, J. (1973). *Spatial autocorrelation*, Pion, London.
- Cliff, A., and Ord, J. (1981). *Spatial processes, models and applications*, Pion, London.
- Cox, W., and Utt, J. (2004). "The Costs of Sprawl Reconsidered: What the Data Really Show. Backgrounder 1770." The Heritage Foundation.
- Cusack, T. R. (1997). "Partisan politics and public finance: Changes in public spending in the industrialized democracies, 1955–1989." *Public Choice*, 91(3–4), 375–395.
- Dahlberg, M., Mörk, E., Rattsø, J., and Ågren, H. (2008). "Using a discontinuous grant rule to identify the effect of grants on local taxes and spending." *J. Public Econ.*, 92(12), 2320–2335.
- De Mello, L. R. (2001). "Fiscal decentralization and borrowing costs: The case of local governments." *Public Finance Rev.*, 29(2), 108–138.
- Dickson, V., and Yu, W. (1997). "Spending by Canadian provincial governments: An empirical analysis." *Public Finance*, 52(2), 145–160.
- Easterly, W., and Levine, R. (1997). "Africa's growth tragedy: Policies and ethnic divisions." *Q. J. Econ.*, 112(4), 1203–1250.
- Fabricant, S. (1952). *The trend of government activity in the United States since 1900*, National Bureau of Economic Research, New York.
- Falch, T., and Tovmo, P. (2003). "Norwegian local public finance in the 1930s." *Eur. Rev. Econ. Hist.*, 7(1), 127–154.
- Farnham, P. G. (1985). "Re-examining local debt limits: A disaggregated analysis." *Southern Econ. J.*, 51(4), 1186–1201.
- Fingleton, B., and López-Bazo, E. (2003). "Explaining the distribution of manufacturing productivity in the EU regions." *European Regional Growth*, B. Fingleton, ed., Springer, Berlin.
- Foucault, M., Madies, T., and Paty, S. (2008). "Public spending interactions and local politics. Empirical evidence from French municipalities." *Public Choice*, 137(1), 57–80.
- Gabe, T. M., and Bell, K. P. (2004). "Tradeoffs between local taxes and government spending as determinants of business location." *J. Reg. Sci.*, 44(1), 21–41.
- Galli, E., and Rossi, S. (2002). "Political budget cycles: the case of the western German Lander." *Public Choice*, 110(3–4), 283–303.
- Grob, U., and Wolter, S. C. (2007). "Demographic change and public education spending: A conflict between young and old?" *Educ. Econ.*, 15(3), 277–292.
- Hagen, T. P., and Vabo, S. I. (2005). "Political characteristics, institutional procedures and fiscal performance: Panel data analyses of Norwegian local governments, 1991–1998." *Eur. J. Polit. Res.*, 44(1), 43–64.
- Hines, J. R., and Richard, H. (1995). "Anomalies: The flypaper effect." *J. Econ. Perspect.*, 9(4), 217–226.
- Holcombe, R. G., and Williams, D. (2008). "The impact of population density on municipal government expenditures." *Public Finance Rev.*, 36(3), 359–372.
- Hoyt, W. H., and Toma, E. F. (1993). "Lobbying expenditures and government output: The NEA and public education." *Southern Econ. J.*, 60(2), 405–417.
- Hulten, C. R., and Peterson, G. E. (1984). "Is the United States underinvesting in public capital? The public capital stock: Needs, trends and performance." *Am. Econ. Rev.*, 74(2), 166–173.
- Imbeau, L. M., Pétry, F., and Lamari, M. (2001). "Left-right party ideology and government policies: A meta-analysis." *Eur. J. Polit. Res.*, 40(1), 1–29.
- Jones, B. D., True, J. L., and Baumgartner, F. R. (1997). "Does incrementalism stem from political consensus or from institutional gridlock?" *Am. J. Polit. Sci.*, 41(4), 1319–1339.
- Kelejian, H., and Robinson, D. (1992). "Spatial autocorrelation: A new computationally simple test with an application to per capita county policy expenditures." *Reg. Sci. Urban Econ.*, 22(3), 317–331.
- Kiewiet, R., and Szalaky, K. (1996). "Constitutional limitations on borrowing: An analysis of state bonded indebtedness." *J. Law Econ. Organ.*, 12(1), 62–97.
- Ladd, H. F. (1992). "Population growth, density and the costs of providing services." *Urban Stud.*, 29(2), 273–295.
- Ladd, H. F. (1994). "Fiscal impacts of local population growth: A conceptual and empirical analysis." *Reg. Sci. Urban Econ.*, 24(6), 661–686.
- Ladd, H. F., and Yinger, J. (1991). *America's ailing cities: Fiscal health and the design of urban policy*, Johns Hopkins University Press, Baltimore, MD.
- Lago-Peñas, I., and Lago-Peñas, S. (2009). "Does the nationalization of party systems affect the composition of public spending?" *Econ. Gov.*, 10(1), 85–98.
- Lago-Peñas, S. (2008). "Local governments' asymmetric reactions to grants: Causes and consequences." *Public Finance Rev.*, 36(2), 219–242.
- Lalvani, M. (2002). "The flypaper effect: Evidence from India." *Public Budgeting Finance*, 22(3), 67–88.
- Lee, G. K. L., and Chan, E. H. W. (2008). "Factors affecting urban renewal in high-density city: Case study of Hong Kong." *J. Urban Plann. Dev.*, 134(3), 140–148.
- Levaghi, R., and Zanola, R. (2003). "Flypaper effect and sluggishness: Evidence from regional health expenditure in Italy." *Int. Tax Public Finance*, 10(5), 535–547.
- Litman, T. (2004). *Evaluating transportation land use impacts*, Victoria Transport Policy Institute, Victoria, BC, Canada.
- Marlow, M. L., and Shiers, A. F. (1999). "Do law enforcement expenditures crowd-out public education expenditures?" *Appl. Econ.*, 31, 255–266.
- Najafi, M., Mohamed, R., Tayebi, A. K., Adelaja, S., and Lake, M. B. (2007). "Fiscal impacts of alternative single-family housing densities." *J. Urban Plann. Dev.*, 133(3), 179–187.
- Oates, W. E. (1969). "The effects of property taxes and local public spending on property values: An empirical study of tax capitalization and the Tiebout hypothesis." *J. Polit. Econ.*, 77(6), 957–971.
- Okun, A. M. (1981). *Prices and quantities*, Brookings Institute, Washington, DC.
- Paelinck, J. H. P., and Klaasen, L. H. (1979). *Spatial econometrics*, Saxon House, Farnborough, UK.
- Pettersson-Lidbom, P. (2001). "An empirical investigation of the strategic use of debt." *J. Polit. Econ.*, 109(3), 570–583.
- Poterba, J. M. (1994). "State responses to fiscal crises: The effects of budgetary institutions and politics." *J. Polit. Econ.*, 102(4), 799–823.
- Poterba, J. M. (1995). "Capital budgets, borrowing rules, and state capital spending." *J. Public Econ.*, 56(2), 165–187.

- Poterba, J. M. (1997). "Demographic structure and the political economy of public education." *J. Policy Anal. Manage.*, 16(1), 48–66.
- Poterba, J. M. (1998). "Demographic change, intergenerational linkages, and public education." *Am. Econ. Rev.*, 88(2), 315–320.
- Rattsø, J., and Tovmo, P. (2002). "Fiscal discipline and asymmetric adjustment of revenues and expenditures: Local government responses to shocks in Denmark." *Public Finance Rev.*, 30(3), 208–234.
- Rivers, M. J., and Yates, B. M. (1997). "City size and geographic segmentation in the municipal bond market." *Q. Rev. Econ. Finance*, 37(3), 633–645.
- Roubini, N., and Sachs, J. D. (1989a). "Political and economic determinants of budget deficits in the industrial democracies." *Eur. Econ. Rev.*, 33(5), 903–938.
- Roubini, N., and Sachs, J. D. (1989b). "Government spending and budget deficits in the industrial countries." *Econ. Policy Eur. Forum*, 4(8), 99–132.
- Seitz, H. (2000). "Fiscal policy, deficits and politics of subnational governments: The case of the German Laender." *Public Choice*, 102(3–4), 183–218.
- Skinner, R. J. (1976). "Technological determinism: A critique of convergence theory." *Comp. Stud. Soc. Hist.*, 18(1), 2–27.
- Solé-Ollé, A., and Sorribas-Navarro, P. (2008). "The effects of partisan alignment on the allocation of intergovernmental transfers. Differences-in-differences estimates for Spain." *J. Public Econ.*, 92(12), 2302–2319.
- Strumpf, K. S. (1998). "A predictive index for the flypaper effect." *J. Public Econ.*, 69(3), 389–412.
- Tellier, G. (2006). "Public expenditures in Canadian provinces: An empirical study of politico-economic interactions." *Public Choice*, 126(3–4), 367–385.
- Thomas, J. C. (1980). "Policy convergence among political parties and societies in developed nations." *Western Polit. Q.*, 33(2), 233–246.
- Tobler, W. (1979). "Cellular Geography." *Philosophy in Geography*, S. Gale and G. Olsson, eds., Reidel, Dordrecht.
- Tovmo, P., and Falch, T. (2002). "The flypaper effect and political strength." *Econ. Gov.*, 3(2), 153–170.
- Travers, T. (2009). "Time for change? Electoral politics after the local and European elections." *Public Policy Res.*, 16(2), 91–96.
- Wagner, A. (1958). "Three extracts on public finance." *Classics in the Theory of Public Finance*, R. Musgrave and A. Peacock, eds., Macmillan, London.